

# WHO Clinical Guidelines for the Treatment of Hypertension

## 1. Diagnosis and Classification

Hypertension is diagnosed when a patient's systolic blood pressure (SBP) is  $\geq 140$  mmHg and/or their diastolic blood pressure (DBP) is  $\geq 90$  mmHg on two separate visits.

- Normal Blood Pressure: SBP  $< 120$  mmHg and DBP  $< 80$  mmHg.
- Prehypertension: SBP 120-139 mmHg or DBP 80-89 mmHg.
- Stage 1 Hypertension: SBP 140-159 mmHg or DBP 90-99 mmHg.
- Stage 2 Hypertension: SBP  $\geq 160$  mmHg or DBP  $\geq 100$  mmHg.
- Hypertensive Crisis: SBP  $> 180$  mmHg and/or DBP  $> 120$  mmHg (requires immediate critical triage).

## 2. Pharmacological Treatment Protocols

First-line pharmacological treatments include:

- ACE Inhibitors (e.g., Lisinopril, 10-40 mg daily) - recommended for patients with diabetes or chronic kidney disease.
- Angiotensin Receptor Blockers (ARBs, e.g., Losartan, 50-100 mg daily).
- Calcium Channel Blockers (CCBs, e.g., Amlodipine, 5-10 mg daily) - highly effective in elderly patients.
- Thiazide Diuretics (e.g., Hydrochlorothiazide, 12.5-25 mg daily).

Beta-blockers (e.g., Metoprolol) are generally reserved for patients with comorbid coronary artery disease or heart failure.

## 3. Lifestyle and Follow-up Recommendations

Lifestyle modifications are mandatory for all hypertension levels:

- Sodium restriction to  $< 2.0$  grams per day.
- DASH (Dietary Approaches to Stop Hypertension) diet, rich in fruits, vegetables, and low-fat dairy.
- Regular aerobic physical activity (at least 30 minutes daily, 5 days a week).
- Limitation of alcohol consumption and cessation of smoking.
- Follow-up: Re-evaluate Stage 1 hypertension within 3-4 weeks; Stage 2 hypertension within 1-2 weeks.